Helix R7 Collaborative Robot Arm - Technical Manual

1. Introduction

The Helix R7 is a six-axis collaborative robot arm designed for machine tending, screw driving and light assembly. Units ship with controller firmware HX-OS 4.2.1. Read the safety chapter before operating the arm.

2. Safety

The emergency stop circuit must be functionally tested every 250 operating hours and after any collision. Never bypass the protective stop inputs. Operators must complete the Level 2 Helix certification course before changing safety parameters.

3. Specifications

Parameter	Value
Maximum payload	7 kg
Reach	912 mm
Pose repeatability	±0.02 mm
Arm weight	24.5 kg
Ingress protection	IP54
Supply voltage	48 V DC

Table 1: Helix R7 specifications

4. Startup

When the controller boots it executes a fixed sequence of steps before any program can run.

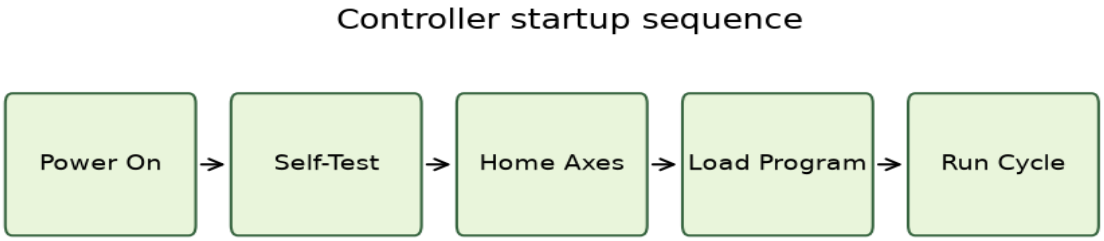

Figure 1: Controller startup sequence

5. Mechanical layout

The arm consists of six revolute joints arranged as shown in Figure 2.

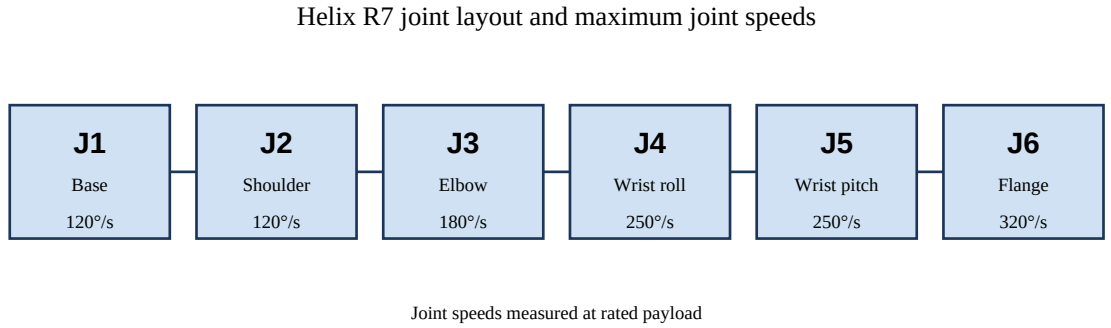

Figure 2: Joint layout of the Helix R7

6. Troubleshooting

The controller reports faults with a three-digit error code.

Error code	Meaning	Corrective action
E101	Overcurrent on joint J2	Reduce payload or speed
E204	Encoder fault	Recalibrate the joint encoder
E305	Communication timeout	Check the EtherCAT cable
E412	Over-temperature	Allow the arm to cool for 30 minutes

Table 2: Controller error codes